FPGA-based mix-precision Monte Carlo using high level synthesis tools

EGH490-1 Project Proposal

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1 General Objective

The primary objective of this project is to explore how reconfigurable computing can be used to reduce the total power consumption of heterogeneous computing environment, such as data centres. We will develop practical implementations of Monte Carlo algorithms to benchmark against theoretical techniques that have previously been explored in the literature. To do this, we will perform a case study utilising field programmable gate arrays (FPGAs) and high-level synthesis tools to understand the feasibility of adopting and utilising reconfigurable devices as accelerator hardware in these heterogeneous frameworks. add to — should i be talking about mixed-precision methods here?

1.1 Research Questions

1. To what extent can implementations of mixed-precision Monte Carlo on reconfigurable computing hardware improve the power efficiency of algorithms compared to graphical processing units?
2. To what extent can high-level synthesis tools and portable code frameworks be used to reduce the development time of algorithms in heterogeneous computing environments¿

2 Key Finding from the Literature

todo after literature is finalised

3 Stakeholders & Resources

When conducting this research we will be considering several different end users who may use or add to the work completed. The first user that we will consider are the potential researchers who will be utilising the algorithms we develop. This is an important consideration as when researchers produce code to run experiments, development time of the code itself is a consideration in their choice of methods. Therefore, we consider the complexity of the code throughout the design cycle of our project to ensure that there is a clear benefit for the additional time cost for these methods in terms of power or time efficiency improvements.

We will also be working closely with the HPC DevOps team throughout the entirety of the project. This team will provide key support to the project in ensuring that progress can be made in the hardware development and application of methods on physical hardware. This collaboration will also provide an opportunity for this work to contribute to the HPC, as we will be assisting in validating software upgrades to the existing FPGA cluster. talk more here?

To be able to complete this project there are several key resources that need to be secured. The most critical of these being access to an FPGA on the QUT high performance computer (HPC) introduce HPC acronym earlier. This involves the coordination of appropriate software installation and hardware access permissions with Hamish Macintosh, DevOps Team Lead with HPC.

4 Project Methodology

4.1 Methods

4.2 Results

5 Deliverables

The expected outcome of this project will be the development and verification of algorithms using high-level synthesis tools for FPGA devices. The main contribution of this work will be benchmarking results and analysis in how FPGAs and reconfigurable computing architectures can be implemented in heterogeneous computing environments to reduce power consumption for compute intensive tasks. This will include a detailed report on the usability of, and development time for complex algorithms

Item	Description	Cost
AMD U55C FPGA	The AMD U55C FPGA is a HPC accelerator card that is currently installed on the QUT HPC, *Aqua*. This FPGA has built-in bootloader and communication protocols which is dissimilar to development FPGAs which most commonly require the engineer to build the entire system from scratch.	\$0.00
Intel CPU	A CPU cluster will be utilised on the HPC as the *host* for running simulation algorithms, where the FPGA or GPU is the compute device. The CPU cluster will also be used for hardware simulation and the generation of bitstream to offload work from personal computers and allow job queues to run consistently.	\$0.00
AMD GPU	We will be utilising an AMD GPU to allow us to perform fair efficiency comparisons between a traditional heterogeneous computing environment and the FPGA accelerated cluster. By choosing AMD hardware for this we are also able to use extremely similar kernel code to minimise the risk of code optimisation being a factor in efficiency results.	\$0.00
Xilinx Vivado Vitis	The Vivado Vitis software tool chain allows for the simulation of hardware components locally. This means we are able to test and validate methods extensively with a much lower time cost before compiling and sending to the FPGA cluster.	TBD

Table 1: List of resources required to complete the describe project.

using, these HLS tools to better understand the feasibility of adopting FPGAs into scientific computing.

Deliverable	Stakeholder	Due Date
EGH490-1 Project Proposal Draft	Supervisors	21/08/26
FPGA Simulation Example Simulation	Supervisors	04/09/26
EGH490-1 Project Proposal	Supervisors	04/09/26
Preliminary Simulation Findings	Supervisors	11/09/26
Geometric Brownian Motion Simulation	Supervisors	02/10/26
EGH490-1 Progress Report Draft	Supervisors	09/10/26
EGH490-1 Progress Report	Supervisors	23/10/26

Table 2: Deliverable outcomes of EGH490-1

Deliverable	Stakeholder	Due Date
EGH490-1 Project Proposal Draft	Supervisors	21/08/26
FPGA Simulation Example Simulation	Supervisors	04/09/26
EGH490-1 Project Proposal	Supervisors	04/09/26
Preliminary Simulation Findings	Supervisors	11/09/26
Geometric Brownian Motion Simulation	Supervisors	02/10/26
EGH490-1 Progress Report Draft	Supervisors	09/10/26
EGH490-1 Progress Report	Supervisors	23/10/26

Table 3: TODO: Deliverable outcomes of EGH490-2

6 Risks, Requirements & Constraints

Since this project has key dependencies that must be met in order to progress, specifically access and functionality of the FPGA on the QUT HPC. If access cannot be gained to this resource or there are issues with the installation of the appropriate software to run HLS tools on the card, implementation of algorithms will be limited. To mitigate this risk, development of the deliverable software will be done in parallel with simulation using the Xilinx Vitas tool-chain which can be used to generate estimated results for the FPGAs performance. Due to the complexity of algorithms being developed using the simulated hardware, the computational bandwidth required is unknown. As such, the QUT HPC will be used for the execution of simulations should it be too intense for a personal computer.

Secondary risks to the project are the compatibility of portable software kernels amongst devices, constrained by the availability of devices on the QUT HPC. A goal of the project is also to investigate the portability of software written in the OpenCL language as a way to reduce development time of algorithms. However, recently AMD have deviated the OpenCL standard to the Heterogeneous-computing Interface for Portability (HIP) framework. To reduce the impact of this on the project, the code base will written in HIP and bench marked on AMD compute device to provide comparable to an OpenCL implementation.

To ensure that we complete this project safely, we have conducted a risk assessment which can be seen in Appendix B.

Through this engineering project, we will dutifully uphold the Engineers Australia Code of Ethics. Since we will be working closely with methods in the literature, to maintain integrity we will verify that our methods appropriately deviate from existing work. The literature that we work with and build upon will be correctly referenced and cited throughout the project. We will practise competently by focusing heavily on the software engineering and computer science aspects of this project in which we have the most expertise. We follow from this with supporting diversity of knowledge through the acknowledgement of areas of less competency. Being advised by and communicating with external engineering practitioners, such as the project supervisors with expertise in electrical engineering and mathematical biology, as well as HPC experts with knowledge in computers science and program optimisation. We later detail how we promote sustainability within our project and practices in the following section.

7 Quality & Sustainability

To ensure that the quality of outcome from this research, we will structure the milestone deliverables such that the codebase and corresponding simulation outputs match existing literature and verified results. This is crucial in ensuring the validity of solutions, as this project aims to work closely with novel implementations of algorithms through the use of compilation pipelines that introduce significant abstraction. This is accompanied by the staged implementation of progressively more complex biological models. By ensuring the validity of algorithms initially on simple models — Geometric Brownian Motion cite — with well understood analytical behaviour we can ascertain whether the algorithms are performing per their intention.

While validating the mathematical accuracy of models is important in ensuring the statistical usability of the research, the main focus of the work will be to explore the power usage of computation is impacted through the utilisation of reconfigurable hardware. Therefore, a key result that we will be looking to understand is the total power consumption of the device for an allotted amount of calculations. It is important to the outcomes of this research that these results are reflective of the conditions that this equipment will be used, i.e. in data centres. Hence, we will be targeting consumption readings directly from the equipment we use on the HPC. This will be accompanied by hardware simulations in preliminary findings to make initial comments on observed performance.

Sustainability is a key focus of this project, specifically we are exploring how reconfigurable devices in heterogeneous computing environments can lower the total power consumption for a HPC with equivalent computational performance. While this is the end goal, we are required to do significant testing on the QUT HPC. As such, it is important to consider the short-term impact this has on the environment through via power utilisation. We aim to mitigate this by ensuring all algorithms are validated through simulation testing before being executed on the HPC.

8 Timeline & Deliverables

The project will be split into three main phases, each with key deliverables that will flow on to the final outcome of the project. The first, and most critical, phase of the project will be making the FPGA cluster on the QUT HPC usable to submit interactive jobs. As previously discussed, this presents the most risk to the project and as such this phase will be accompanied with similar work on simulated hardware models. The goal of which being the execution of *Hello World* examples on hardware to develop understanding of the processes required to build more complex algorithms. talk about when this will be achieved

Following the success of this first phase will be when the main body of work is completed. This will be the creation of both simple and complex simulation algorithms, described in the literature, on the available FPGA card. To mitigate potential risks in the usage of the physical hardware on the HPC, simulation models will first be created. This mitigation is twofold as the creation of FPGA firmware is extremely time intensive, the simulation allows us to experiment with a lower time-cost model before transferring to hardware. Key deliverables in this phase are simulation algorithms written in the AMD portable language, as well as validity of outcome tests and preliminary efficiency measures. talk about when this will be achieved

To complete the work, the final phase of the project will be an in-depth analysis of the power efficiency of the simulation algorithms on the FPGA, compared to a CPU or GPU. This will be complemented by the specific programming language used, which will allow us to use almost identical code on each device to conduct fair analysis.

Number	Focus	Deliverable	Dependency	Milestone
1	Literature Review	Literature review report	NA	TBD
2	Preliminary FPGA on HPC usage	TBD	NA	TBD
2.1	Vivado Vitis example code	TBD	NA	TBD
2.2		TBD	NA	TBD

Table 4: TODO: Timeline of deliverable outcomes for EGH490

9 Management of Project Changes

Due to the several layers of dependency of external factors for this project to be completed, we have formulated the goals of the work to be flexible. Meaning that if we are unable to gain access to physical hardware on the HPC within a reasonable time frame to implement, and experiment with, the algorithms we will conduct the research only on simulated hardware. This project is also entirely academic with weekly meetings with the supervising staff and constant communication available. As such, should a request be made for the project, or issues arise from other dependencies, we will be able to quickly pivot the methodology of the research to continue relevant work. These changes will also be succinctly communicated to the supervisors of the project through email and discussed at the weekly meeting.

Sign off

Name	Signature	Date
Luca Lubrano	LL	04/09/2026

A Literature Review

Through this literature review we aim to summarise the key methodologies in mixed-precision Monte Carlo methods in the literature that are relevant to our work. In this report we work to understand how we can utilise and exploit reconfigurable hardware with mixed-precision to take advantage of unique features in the computing architectures of FPGAs over GPUs. We perform this research within the context of Mathematical Biology in which large quantities of statistical independent simulations are required to produce meaningful results. This presents a clear use case for these methods as there is potential for both the wall-time and power efficiency improvements over conventional computing frameworks.

A.1 Heterogeneous Computing - reword

The paper by Haas and Giles [1] focuses on the application of these methods to Field Programmable Gate Arrays (FPGAs) — devices that can be programmed to the logic gate level. This means that a user can define entirely custom bit-width variable types. An extremely beneficial tool in the case of the mixed precision multilevel scheme, as the further we can reduce the bit length in lower levels, the cheaper and hence more simulations we are able to use, reducing the variance of the estimator for an equivalent cost.

The numerical models — mainly geometric Brownian motion — that Haas and Giles [1] use to validate their methods are relatively simple, by the fact they can be broken down into their elementary binary operations. In the context of Systems Biology however, this is often not possible as realistic models are too complex to perform this kind of decomposition. As such, we are unable to explore the more detailed intricacies of the proposed bit-width optimisations in this work.

Instead, we focus on the more commonly available bit-widths on modern CPU and GPU chips. To maximise the clock cycle efficiency in our computing, we wish to keep our computations to the size of the floating point registers available on the target hardware. For current GPUs, this is double 64-bit, single 32-bit, and half 16-bit. Working with these register sizes is critical in the efficiency of our methods. By doing this, we can increase the throughput of our algorithm per clock cycle by using *Single Instruction Multiple Data* (SIMD) operations [2, 3].

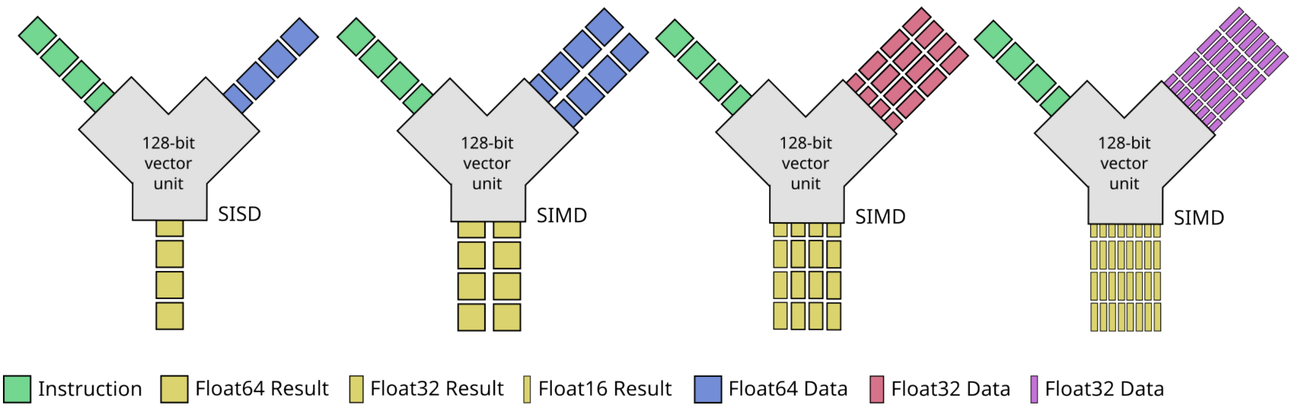

Figure 1: Schematic of vector unit operations in single instruction single data and single instruction multiple data configurations with varying precision levels. needs to be fixed

This works as the register is of a fixed size, usually 64-bit, each clock cycle then pushes an instruction to the register where it completes the operation on the data held. If we reduce the bit width of our data to a less precise variable type, i.e. 32 bit, the register is able to perform the same operation on two independent sets of data [4–6], see diagram in Figure 1. Importantly, it is able to do this in the same clock cycle as if the data had been a single full precision width. This is a key part in how mixed-precision methods can provide significant speed up in high performance computing in lower precision levels [7].

Kimpson et al. [8] investigate methods for computing simulation estimates using mixed and reduced precision in the context of SSAs. These algorithms aim to minimise the error in approximation while using native 16-bit arithmetic. At a hardware-level, this reduces the memory footprint while increasing the SIMD operations able to be computed. They specifically target GPU and tensor processing unit (TPU), which provides insight into how methods can be mapped between device hardware and the results they can achieve in heterogeneous environments.

A key advantage provided by conducting our computations in a heterogeneous environment is that we are able to optimise the choice of device based on suitability to the task [9–11]. Meaning we are able to maximise the utilisation of hardware, which is beneficial since general purpose compute devices, such as CPUs and GPUs, carry high idle power costs. These systems often use a variety of these general purpose devices in combination with accelerator hardware, including application-specific integrated circuits (ASIC), intelligence processing units (IPU) and FPGAs [12].

A.1.1 Reconfigurable Computing - reword

Reconfigurable computing is classified by a computer architecture that is able to change its logic programming. The most common and popular example of which are FPGAs, built from 'reconfigurable fabric' which contains the configurable logic blocks, interconnections and BRAM that allow the hardware to change its logic functions [12]. In computing, this means that we are able to construct programs not at a software level, but instead program them onto the physical architecture of the card. This is a key aspect of the power efficiency that these devices can provide. Unlike CPUs and GPUs, they do not have any overhead power cost related to unnecessary and always active processes.

As a result of the freedom provided by FPGA hardware, we are also able to increase computational efficiency in low precision settings through the creation of unique precision levels. Unlike modern CPUs which can provide either double (64-bit), float (32-bit) or half-precision (16-bit), we are able to define custom bit length precision data types. This can be directly applied in areas such as multi-fidelity Monte-Carlo simulation, which sample cheap, low-precision simulations for Bayesian inference acceleration.

These devices have been traditionally used by electrical engineers for hardware prototyping and development, requiring users to understand low-level hardware description languages (HDLs). Making them inaccessible to scientific researchers without the proper background to fully take advantage of their utility. Recent advancements in compilation abilities have lead to the ability for OpenCL to be compiled into bit-stream and hence are now able to be used to program these devices. This has made them a target of active research in areas such as high-dimensional linear algebra [3, 5, 9, 11, 13], simulation problems [4] and time-critical safety operations [2].

Intelligence processing units (IPUs) are an emerging technology driven by the need for more specialised Artificial Intelligence computing capabilities [14]. Unlike traditional GPU architecture which uses single-instruction multiple data (SIMD) vectorised parallelism to accelerate tasks, the IPU utilises multi-instruction multiple data (MIMD) [15]. This allows each thread to perform operations in a distinct flow, and irregular data access patterns without loss of efficiency [15]. Similarly to FPGAs, these devices are also able to define custom precision data types, which has made them extremely efficient processors in the AI inference problem [16–18].

A.2 Stochastic Simulation

similar to MXB372

A.3 Monte Carlo Methods - reword

Monte Carlo methods are a common first step in understanding the average behaviour of observed system states given some known parameter inputs [19, 20]. If one were to want to know the expectation of a function P at some time t given an observed state S it is intuitive to think that we can take repeated samples of the system to determine an average behaviour [21]. Formally, to approximate $\mathbb{E}[P]$ a Monte Carlo estimator is the average values $P(\mathbf{X}(t))$ for N i.i.d. samples $\mathbf{X}(t)$

$$\mathbb{E}[P(\mathbf{X}(t))] = \int P(\mathbf{x})p(\mathbf{x})\mathrm{d}x,$$

$$\lim_{N\to\infty} \sum_{n=1}^{N} P(\mathbf{X}(t)^{(n)}) = \mathbb{E}[P(\mathbf{X}(t))],$$

where $p(\mathbf{x})$ is the probability density function of the random variable $\mathbf{X}(t)$. With variance $V_N = \frac{1}{N}\mathbb{V}[P]$, therefore, in order to achieve a root mean squared error (RMSE) of ϵ we require $N = \mathcal{O}(\epsilon^{-2})$. This is where the Monte Carlo method becomes computationally unwieldy, as each of these simulations often involves the solution to an Itō stochastic differential equation (SDE) or a costly approximation.

$$X_i(t) = X_i(0) + \sum_{j=1}^{M} \nu_{i,j} \int_0^t a_j(\mathbf{X}(\eta(s)); \boldsymbol{\theta})\, ds + \sum_{j=1}^{M} \nu_{ij} \int_0^t \sqrt{a_j(\mathbf{X}(\eta(s)); \boldsymbol{\theta})}\, dW_j(s), \tag{1}$$

noting that $X_i(t)$ is the value of a random solution path at some time $t > 0$. When computing these estimates, we have a statistical error from the Monte Carlo and a bias introduced by the discretisation of the Itō SDE, thus we have,

$$\text{MSE} = \text{Var} + \text{Bias}^2.$$

If we want a RMSE less than ϵ we need $N = \mathcal{O}(\epsilon^{-2})$. However, we also need $\tau = \mathcal{O}(\epsilon)$ since Bias $= \mathcal{O}(\tau)$ for Euler Maruyama. As a result, the total cost of the estimator is $\mathcal{O}(\epsilon^{-3})$.

Due to this, we seek methods for generating more accurate results while minimising the number of Monte Carlo samples we need to take. Methods such as quasi-Monte Carlo in which each sample is chosen deterministically to reduce the error [22]. An alternative, the multilevel Monte Carlo method, uses a coupling scheme to apply *control variate reduction* to minimise simulation cost for some desired RMSE ϵ.

A.4 Mixed-precision Methods

similar to MXB372

B Risk Assessment

Risk	Likelihood	Consequence	Control type	Control
Musculoskeletal strain/poor posture from prolonged sitting and computer use	Medium	Low	Administrative	Take regular breaks (15 minutes per hour), maintain correct posture, use an adjustable chair and monitor height in line to be in line with eye level with elbows at 90 degrees.
Eye strain and mental fatigue from extended screen exposure	Medium	Low	Administrative	Follow the 20-20-20 rule (every 20 minutes, look at something 20 feet away for 20 seconds), adjust screen brightness/contrast, ensure adequate ambient lighting
Loss of thesis work, code, or data due to hardware failure, corruption, or accidental deletion	Medium	High	Engineering & Administrative	Use git version control with a remote repository, push commits to remote repository daily
Electrical or trip hazard from computer equipment, chargers, and loose cabling	Low	Medium	Engineering	Ensure all equipment is electrical tested where required, route cables away from walkways, keep the desk area tidy and free of clutter
Burnout or stress from extended, isolated independent work under project deadlines	Medium	Medium	Administrative	Set realistic milestones and a project schedule, maintain regular check-ins with supervisors, schedule rest days and enforce reasonable working hours

Table 5: Project risk assessment

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